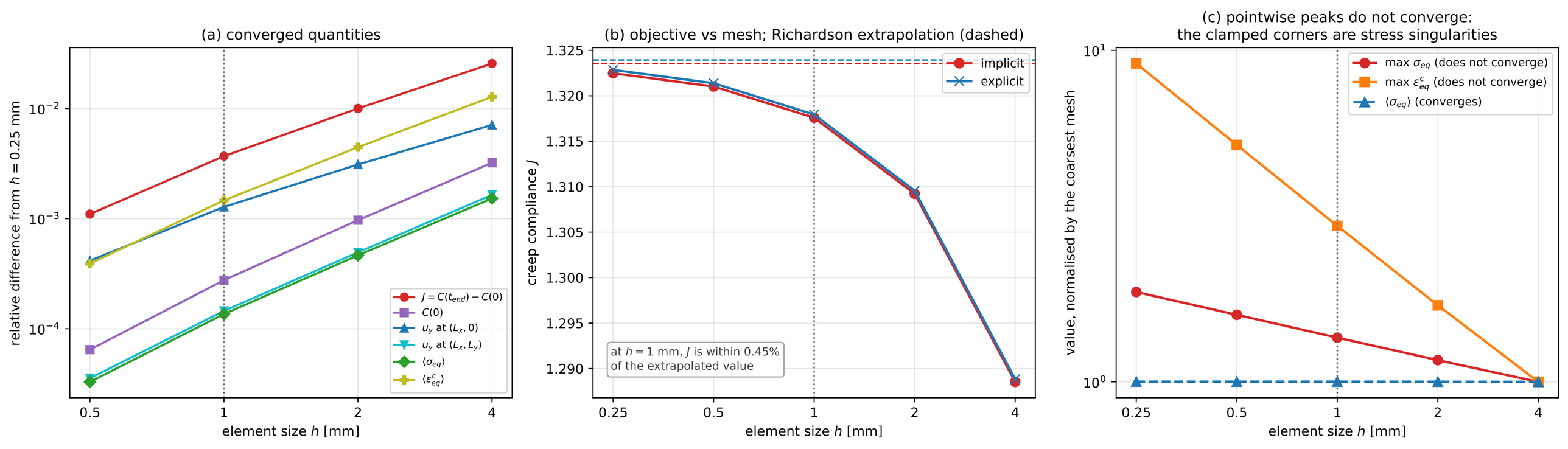
(a) converged quantities
relative difference from $h = 0.25$ mm
10^{-2}
10^{-3}
10^{-4}
0.5
1
2
4
element size h [mm]
$J = C(t_{end}) - C(0)$
$C(0)$
u_y at $\langle L_x, 0)$
u_y at $\langle L_x, L_y)$
$\langle \sigma_{eq} \rangle$
$\langle \varepsilon^c_{eq} \rangle$
(b) objective vs mesh; Richardson extrapolation (dashed)
creep compliance J
1.325
1.320
1.315
1.310
1.305
1.300
1.295
1.290
0.25
0.5
1
2
4
element size h [mm]
implicit
explicit
at $h = 1$ mm, J is within 0.45%
of the extrapolated value
(c) pointwise peaks do not converge:
the clamped corners are stress singularities
value, normalised by the coarsest mesh
10^1
10^0
0.25
0.5
1
2
4
element size h [mm]
max σ_{eq} (does not converge)
max ε^c_{eq} (does not converge)
$\langle \sigma_{eq} \rangle$ (converges)